

College of Engineering
Department of Chemical Engineering



Literature Review
CHE 450 (001), Liquid Rocketry Lab Advanced Projects (LRL AP)

Jose Barbeito

The purpose of this document is to summarize the efforts made by other teams in the space specifically concerning thermals of RDE systems. The three main topics covered are cooling the RDE, simulating and analyzing the system, and the implications of results.

To begin, the primary challenge with operating RDE's at stable operating conditions for any length of time is the high temperatures seen by the exhaust. Though highly dependent on fuel selection, the majority of RDE's produce exhaust gasses at a range of 1000 K to 3000 K, with select models reaching 4000+ K. Due to manufacturing materials, most components downstream of an RDE operating at 1500K+ would melt after several seconds of burn time. There exists two options: cooling the air after it exits the combustion chamber or cooling the combustion chamber directly.

Cooling the RDE can be done in many different ways. The most common and primitive method is to not cool the system at all, simply limiting burn times to a second or less, leaving the system time to cool to ambient temperature between runs. This of course limits the length of runs and does not allow for observation at steady state, though has been done successfully for a longer series of test runs by ^{1 2}

Moving into active cooling, two primary methods exist: cooling the chamber itself and cooling the exhaust. Water cooling the combustion chamber has promising literature, with many teams extending runs and reaching a pseudo steady state. ^{3 4} Smallwood et. al. had runs exceeding 50 seconds using the water cooling method, seeing inner body and outer body wall temperatures of 300 and 250 degrees Fahrenheit, respectively.

Other than cooling the chamber itself, cooling the air using cold ambient air is also viable to ensure survival of downstream components. Braun et al. utilized film cooling to both keep the hot exhaust from contacting the chamber walls and introducing colder air, dropping temperatures down from 2000K to 1500K. ⁵ A similar setup was used by Joest et. al in 2021, utilizing this film cooling on both the inner and outer layers of the RDC. ⁶ Interestingly, adding subsonic colder air could have more potential for our application as the flow would be both colder and slower, leading to higher efficiencies in the downstream turbine.

For simulation and thermal analysis, three main sections should be analyzed: the combustion chamber, the plenum, and the turbine itself. Several teams have used simulations to estimate thermal properties of the system, proving its efficacy in understanding the

¹ Design and optimization of supersonic turbines for detonation combustors - Sousa et. al.

² Design, Buildup, and Testing of a Radial RDE for Compact Aux Power Unit - Huff, Riley

³ Thermal and Structural Characterization of a Rotating Detonation Rocket Engine - Smallwood

⁴ Continuous Spin Detonations - Bykovskii et. al.

⁵ Design, Analysis, and Test of a Cooled Rotating Detonation Combustor - Braun et. al.

⁶ Design of a turbine stage for a rotating detonation combustion test facility - Joest et. al.

thermodynamics within the system.^{7 8 9 10 11 12 13} While using complete CFD programs such as ansys for full-system simulations is possible, several equations can be used to estimate the heat output of the RDE and approximate heat transfer between the gas flow and objects downstream of the combustion chamber.

It should be said that the results of thermal analysis on the RDC and turbine will directly result in design changes and operational guidelines. Using the heat flux from the exhaust and the material for the turbine, we can estimate the maximum allowable time the turbine can be run without causing irreversible blade damage or damage to other auxiliary systems, in addition to keeping the RDC from melting. Being able to accurately simulate a thermal environment is crucial to understanding and preventing key failure modes and will lead to a better designed RDE.

⁷ Thermal power plant upgrade via a rotating detonation combustor and retrofitted turbine with optimized endwalls - Braun et. al.

⁸ Thermodynamic analysis of a gas turbine engine with a rotating detonation combustor - Sousa et. al.

⁹ Pressure Gain Combustion for Land Based Power Generation - Ferguson et. al.

¹⁰ Design and optimization of supersonic turbines for detonation combustors - Sousa et. al.

¹¹ Design, Buildup, and Testing of a Radial RDE for Compact Aux Power Unit - Huff, Riley

¹² Thermal and Structural Characterization of a Rotating Detonation Rocket Engine - Smallwood

¹³ Continuous Spin Detonations - Bykovskii et. al.